

Madukuri Shaurya

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Education

IIT Delhi *B.Tech in Electrical Engineering – CGPA 7.77*

Nov 2020 – May 2024

Experience

Data Scientist II *Skan.AI*

March 2026 – Present

Project: Computer Use Agent (CUA) — Desktop GUI Automation for BFSI.

- **Leading a team of 3** building a **Computer Use Agent** for automated desktop GUI interaction in BFSI workflows.
- Architected a **stateless step executor** driven by an Agentic Orchestration Procedure (AOP) — receives a screenshot + step name, executes atomic GUI actions, returns control.
- Implemented **Set-of-Mark (SoM)** prompting with a thinker-grounder model split for visual grounding.
- Designed **HITL escalation** and grace-condition handling for UI collisions and post-action verification.

Data Scientist I *Skan.AI*

July 2024 – April 2026

Project: Agentic-AI, Robotic Process Automation Discovery Framework.

- Modeled a **next activity prediction** model using a custom-built **transformer** framework.
- Fine-tuned openly available **LLM architectures** like Qwen, Flan-T5, and Gemma.
- Built rolling **automatability scores** at the task sub-sequence level using **perplexity scores**, enabling clients to vary frequency factor and pattern length in process discovery.

Project: Agentic-AI, Automatic Playbook Discovery and Agentic Execution Framework.

- Designed and built an **end-to-end process understanding pipeline** for automatic playbook discovery from raw event logs and user interaction data.
- Leveraged **Ollama-based** open-source models and implemented a multi-model **Retrieval-Augmented Generation (RAG)** system for contextual reasoning and action planning.
- Orchestrated **agentic execution workflows** integrating LLM reasoning with structured process intelligence.
- Productionized the pipeline with optimized **memory utilization** and **latency-aware architecture**, improving inference efficiency and scalability.

Data Scientist Intern *Skan.AI*

May 2023 – Jul 2023

Project: Event Data Extraction from Communication using Large Language Models.

- Developed fine-tuned models (**BERT, BART, Pegasus**) for event data extraction using the **Teacher-Student** methodology.
- Designed custom-built **domain-specific models**, creating compact versions tailored to specialized tasks.
- Applied **clustering algorithms** to improve consistency, ensuring more accurate event data extraction.

Publications

Glaucoma Abnormality Classification using YOLO and SWIN Transformer
Madukuri Shaurya, Anumeha Varma, Monika Agrawal

May 2024

[10.1109/ISBI56570.2024.10635301](https://doi.org/10.1109/ISBI56570.2024.10635301) [🔗](#)

Projects

PaCT: Parallel Continuous Thought for Reasoning in LLMs

- Proposed **PaCT**, a parallel latent reasoning framework that eliminates **recurrent unrolling** in continuous thought by organizing reasoning into stage-wise parallel latent resynthesis.
- Synthesizes latent states via **learnable query codebook + cross-attention**, refined through **bidirectional within-stage interaction**, preserving causal decoding.
- Achieved **4–9× training speedup** over Coconut while matching or improving accuracy on GSM8k, Pron-toQA, and ProsQA; scaled to **LLaMA-3B/8B** with consistent 5× reduction.

Forge — Custom GPU Kernels for LLM Training Optimization

- Built custom **Triton GPU kernels** for memory-efficient LLM training.
- Implemented **fused linear + cross-entropy** kernel with tiled vocabulary computation, avoiding full logit materialization.
- Developed fused **RMSNorm/LayerNorm** kernels — single-pass reduction + normalization.
- Wrote custom **SwiGLU** and **RoPE** kernels with fused forward and backward passes.

DS Coin, Cryptocurrency Development

- Designed and developed a decentralized cryptocurrency model in Java, incorporating **authenticated linked lists**, **Merkle trees**, and various algorithms to ensure secure token storage, transaction validation, and fraud detection.

Fully Homomorphic Encryption Scheme

- Researched and developed advanced cryptographic techniques, including **lattice-based short integer solutions**, **Regev’s encryption scheme** utilizing learning with errors (LWE) and a **fully homomorphic encryption** method based on ring LWE.

Additional Experience

Data Analyst Intern *Jarvis Labs*

Jan 2023 – May 2023

Built **trading analytics** metrics (velocity, liquidity ratio, Gini) using **blockchain data** from Etherscan and Glassnode.

Data and Software Engineer Intern *Fynstack*

June 2022 – Jan 2023

Built backend on **AWS** (GraphQL, AppSync) and implemented **fuzzy merge**/regex filtering to consolidate annual statements.

Skills

Languages: Python, C++, CUDA, C, Java, SQL, React

Tools and Frameworks: PyTorch, Triton, LangChain, LangGraph, MongoDB, Postman, Docker, GraphQL, TensorFlow